MCQS

CS310 midterm preparation file

ZB

بسم الله الرحمن الرحيم

In the name of God , Most Gracious, Most merciful



CS310

MCQS

1. Which of the following is the best description of an algorithm?

A) A process that always terminates

B) A well-defined computational procedure

C) A set of instructions executed in parallel

D) A mathematical model

Answer: B

2. Which data structure is used to implement recursive function calls?

A) Queue

B) Stack

C) Array

D) Linked List

Answer: B

3. Which of the following time complexities is the best for searching in a sorted array?

A) O(n)

B) O(log n)

C) O(n log n)

D) O(1)

Answer: B

4. What is the worst-case time complexity of quicksort?

A) O(n^2)

B) O(n log n)

C) O(n)

D) O(log n)

Answer: A

5. Which of the following is a characteristic of a greedy algorithm?

A) It guarantees the optimal solution

B) It solves the problem step-by-step, considering only the current choice

C) It always produces a correct result

D) It solves problems that require backtracking

Answer: B

6. Which data structure is used to implement the breadth-first search (BFS) algorithm?

A) Queue

B) Stack

C) Linked List

D) Priority Queue

Answer: A

7. Which of the following is the time complexity of binary search in a sorted array?

A) O(n)

B) O(log n)

C) O(n log n)

D) O(1)

Answer: B

8. Which of the following sorting algorithms has the best average case time complexity?

A) QuickSort

B) MergeSort

C) BubbleSort

D) InsertionSort

Answer: A

9. Which of the following operations can be performed using a priority queue?

A) Find the minimum element

B) Find the maximum element

C) Insert an element

D) All of the above

Answer: D

10. Which data structure is best for implementing a recursive function?

A) Queue

B) Stack

C) Linked List

D) Tree

Answer: B

11. Which of the following is an example of a divide-and-conquer algorithm?

A) QuickSort

B) MergeSort

C) Binary Search

D) All of the above

Answer: D

12. What is the worst-case time complexity of insertion sort?

A) O(n log n)

B) O(n^2)

C) O(n)

D) O(log n)

Answer: B

13. Which of the following is true about a doubly linked list?

A) It requires more memory than a singly linked list

B) It only allows traversal in one direction

C) It is slower for insertion and deletion

D) It doesn't allow circular traversal

Answer: A

14. What is the time complexity of finding an element in a hash table?

A) O(n)

B) O(log n)

C) O(1) on average

D) O(n^2)

Answer: C

15. In which type of graph traversal are vertices visited layer by layer?

A) Depth-First Search (DFS)

B) Breadth-First Search (BFS)

C) Topological Sort

D) None of the above

Answer: B

16. Which of the following is NOT a property of a binary search tree (BST)?

A) The left subtree of a node contains only nodes with values less than the node's value

B) The right subtree of a node contains only nodes with values greater than the node's value

C) The height of the tree must always be balanced

D) It allows fast searching

Answer: C

17. Which of the following is the primary characteristic of a dynamic programming approach?

A) It uses recursion to find solutions

B) It solves problems by breaking them into subproblems and storing the results

C) It uses greedy techniques to find the optimal solution

D) It avoids overlapping subproblems

Answer: B

18. Which of the following sorting algorithms is not comparison-based?

A) QuickSort

B) MergeSort

C) RadixSort

D) BubbleSort

Answer: C

19. Which of the following is the correct representation of an adjacency matrix for a graph?

A) A 2D array used to represent which nodes are connected to each other

B) A linked list of edges

C) A collection of vertices only

D) A tree-like structure of vertices

Answer: A

20. What is the time complexity of performing a push operation in a stack?

A) O(n)

B) O(log n)

C) O(1)

D) O(n^2)

Answer: C

21. Which of the following is true about a tree data structure?

A) It is a linear data structure

B) Each node has only one child

C) It is a hierarchical data structure

D) It does not support recursive functions

Answer: C

22. Which of the following algorithms uses a min-heap?

A) Dijkstra's Algorithm

B) QuickSort

C) Binary Search

D) MergeSort

Answer: A

23. Which of the following is the best description of a hash function?

A) It maps data of arbitrary size to a fixed-size value

B) It guarantees unique values for every input

C) It is used only for sorting elements

D) It always produces a sorted array

Answer: A

24. Which of the following operations can be done in O(1) time on a doubly linked list?

A) Insertion at the head

B) Searching for an element

C) Deletion from the middle

D) None of the above

Answer: A

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25. Which of the following is a characteristic of the Floyd-Warshall algorithm?

A) It computes the shortest path between all pairs of vertices

B) It computes the shortest path from a single source to all other vertices

C) It is a greedy algorithm

D) It uses a stack for its implementation

Answer: A

26. Which of the following is a disadvantage of using a linked list over an array?

A) Accessing an element is slower

B) It requires less memory

C) Insertion and deletion are more complicated

D) It cannot be used for recursion

Answer: A

27. Which of the following traversal methods is used in a depth-first search (DFS) algorithm?

A) Pre-order traversal

B) Post-order traversal

C) In-order traversal

D) All of the above

Answer: D

28. What is the space complexity of the MergeSort algorithm?

A) O(1)

B) O(n)

C) O(log n)

D) O(n log n)

Answer: B

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29. Which data structure is used in the implementation of Dijkstra's shortest path algorithm?

A) Priority Queue

B) Stack

C) Queue

D) Hash Map

Answer: A

30. Which of the following graph traversal algorithms can be implemented using recursion?

A) Depth-First Search (DFS)

B) Breadth-First Search (BFS)

C) Both A and B

D) None of the above

Answer: A

31. Which of the following is the time complexity of accessing an element in a linked list?

A) O(1)

B) O(log n)

C) O(n)

D) O(n^2)

Answer: C

32. Which of the following is the space complexity of an algorithm that uses dynamic programming to solve a problem with an array of size n?

A) O(1)

B) O(n)

C) O(n^2)

D) O(log n)

Answer: B

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33. Which of the following is the best description of a heap?

A) A balanced binary tree where the left child is always smaller than the right child

B) A tree-based data structure that satisfies the heap property

C) A queue with priority ordering

D) A linear data structure that allows insertion and deletion

Answer: B

34. Which of the following algorithms is used for finding the shortest path in a graph with weighted edges?

A) Breadth-First Search

B) Dijkstra's Algorithm

C) Depth-First Search

D) Bellman-Ford Algorithm

Answer: B

35. What is the space complexity of the quicksort algorithm?

A) O(1)

B) O(n)

C) O(n log n)

D) O(log n)

Answer: D

36. Which of the following algorithms is NOT used for graph traversal?

A) Depth-First Search (DFS)

B) Breadth-First Search (BFS)

C) MergeSort

D) Dijkstra's Algorithm

Answer: C

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37. Which of the following is true for an AVL tree?

A) It is a type of binary search tree where the balance factor of each node is either 0, 1, or -1.

B) It is an unbalanced binary search tree.

C) It has a fixed height for all nodes.

D) It is a type of binary tree that does not allow duplicates.

Answer: A

38. What is the time complexity of finding the maximum element in a heap?

A) O(1)

B) O(log n)

C) O(n)

D) O(n log n)

Answer: A

39. Which of the following algorithms is used to find the minimum spanning tree of a graph?

A) Kruskal's Algorithm

B) Dijkstra's Algorithm

C) Floyd-Warshall Algorithm

D) Bellman-Ford Algorithm

Answer: A

40. What is the worst-case time complexity of a binary search tree (BST) when it is unbalanced?

A) O(log n)

B) O(n log n)

C) O(n)

D) O(n^2)

Answer: C

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41. Which of the following is true about a graph with a directed edge between two vertices?

A) The graph is undirected

B) The edge has a direction from one vertex to another

C) The graph contains no loops

D) The graph cannot have cycles

Answer: B

42. Which of the following is an example of a non-comparison-based sorting algorithm?

A) MergeSort

B) HeapSort

C) QuickSort

D) CountingSort

Answer: D

43. Which of the following operations can be performed in O(1) time on a stack?

A) Push

B) Pop

C) Peek

D) All of the above

Answer: D

44. Which of the following is the time complexity of bubble sort in the best case scenario?

A) O(n log n)

B) O(n^2)

C) O(n)

D) O(log n)

Answer: C

45. Which of the following graph algorithms can be used to detect negative weight cycles in a graph?

A) Dijkstra's Algorithm

B) Bellman-Ford Algorithm

C) Depth-First Search

D) Prim's Algorithm

Answer: B

46. Which data structure is used in the implementation of the depth-first search (DFS) algorithm?

A) Queue

B) Stack

C) Linked List

D) Binary Tree

Answer: B

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channel on WhatsApp:

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Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.
Therefore, for passing a course/subject, student shall fulfill the following:
a) Secure minimum 20% score in Formative Assessments/Mid terms
b) Secure minimum 20% score in Final Term Examinations.
c) Secure at least 40% marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?
A: Follow these steps:
1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).
2. Each subject typically has 3 credit hours.
3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.
Examples:
21 credit hours = 7 su18 credit hours = 6 subjects
15 credit hours = 5 subjects
Remember, the university may allow different credit hours for each student, so check your specific allowance.
By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

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Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed
Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat Fame

Moat

Humy pta hona chaheay Moat tab ani ha jab Allah chahy us se pehly puri dunya bi ak tararf ho ke apko marna chahy to apka kuch nahe beggar saqti

Risk/Dolat

Risk Dolat sb Allah pak ke hath m ha puri dunya ak taraf b ho jay na apsy apka risk cheen saqti ha na de saqti ha Ye sirf ALLAh pak ky hath m ha

Izat Zilat shohrat Fame

Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b apko zra brabr b demage nahe kr saqti Agr ALLAH

Apko izat dena chahay to puri dunya ak tararf ho ke b apko 1% b nuksan nahe pohancha saqti
Phr Dar ks bat ka ??????????????
Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne mat jhuko puri dunya mel k b apka kuch b nahe beggar saqti

ZB
MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family
Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my compaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha
Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us
m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us
Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

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Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

MY Family

ust Relax and focus! Exams are not difficult. Put in your 100% effort and trust in Allah.

AL-Quran

"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"

May Allah bless you